SeedSelect: Hybrid Test-Time Verification for Inference-Time Compute Scaling in Single-Image 3D Reconstruction

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Abstract. Inference-time compute scaling—generating multiple candidates and selecting the best—has emerged as a powerful paradigm in language and image generation, but remains underexplored for 3D reconstruction. We study this setting in diffusion-based single-image 3D pipelines and identify a strong *seed lottery*: different random seeds produce reconstructions whose geometric quality can vary by multiple times for the same input. We introduce *SeedSelect*, a test-time selection framework with two verifiers: (i) a zero-shot multi-view consistency score using Difix3D+ as a 2D refinement-based proxy, and (ii) a learned reranker. We further propose a *hybrid* selector that combines zero-shot coarse filtering with learned fine-grained reranking under risk-aware utility. On GSO-300 (K=4), zero-shot closes 22.1% of oracle gap, learned-only closes 68.9%, and hybrid closes 42.3% at substantially lower learned inference cost. On OmniObject3D-100, hybrid closes 52.3% of oracle gap, outperforming both zero-shot (43.7%) and learned-only (42.0%) under domain shift. Across settings, the hybrid policy improves quality-cost trade-off and maintains robust cross-domain behavior. These results establish a practical and scalable route for inference-time compute scaling in 3D reconstruction.

Keywords: single-image 3D reconstruction · inference-time compute scaling · quality estimation · hybrid verification

1 Introduction

Inference-time compute scaling—the practice of allocating additional computation at test time to improve output quality—has emerged as a transformative paradigm across machine learning. In language modeling, generating multiple candidate responses and selecting via reward models consistently outperforms single-shot generation [20, 21]. In 2D image generation, Ma *et al.* [11] recently demonstrated that scaling inference-time compute for diffusion models can be more effective than scaling model parameters. Yet this paradigm remains underexplored for single-image 3D reconstruction, despite the fact that diffusion-based image-to-3D pipelines are inherently stochastic and exhibit substantial quality variation across random seeds.

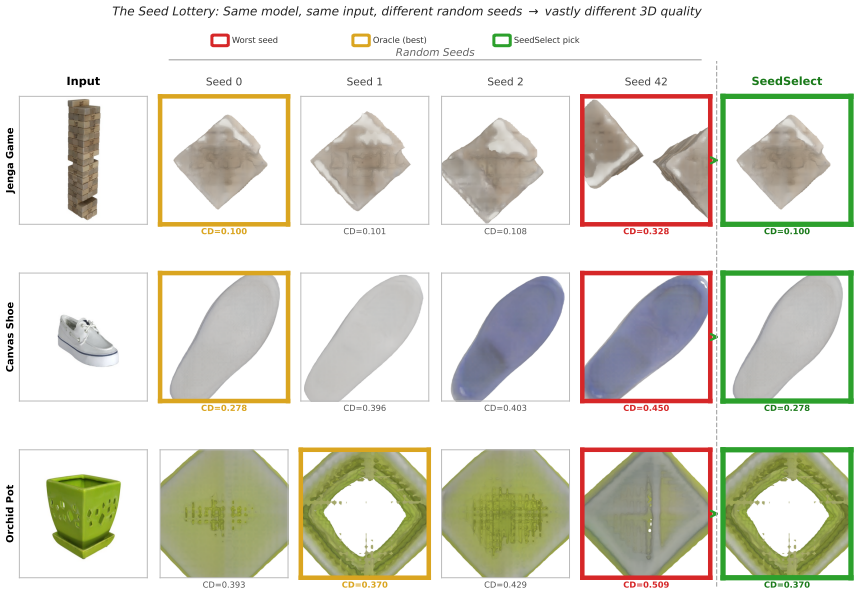

Fig. 1: Inference-time compute scaling for 3D reconstruction. Diffusion-based pipelines produce reconstructions of vastly different quality depending on the random seed. SeedSelect identifies the best candidate using Difix3D+ [30] as a zero-shot verifier: *less 2D refinement needed ⇒ better 3D quality.*

We systematically investigate this opportunity, revealing what we term the *seed lottery*: running the same reconstruction pipeline with different random seeds yields Chamfer Distance variations of up to 3.3× for the same input. Simply selecting the best of K=4 candidates with an oracle achieves 6.6% improvement on GSO [2] and 11.9% on OmniObject3D [27], and these gains increase to 14.3% with K=16 candidates. This reveals a large pool of "free" quality hidden behind the randomness of diffusion sampling, available to any method that can identify the best candidate.

The core challenge is designing a *verifier*—a function that can rank candidate reconstructions without ground truth. Standard mesh quality metrics (face regularity, smoothness) show no correlation with true geometric accuracy, and PSNR-based scoring fails due to systematic camera and lighting mismatches. We show that even strong vision encoders (CLIP, DINOv2) are unreliable in this setting, motivating a domain-specific verification design.

We propose **SeedSelect**, a test-time verification framework with a strong zero-shot verifier and a hybrid extension. Our key insight is that *a 3D reconstruction requiring less 2D refinement across rendered views is typically geometrically better*. We operationalize this using Difix3D+ [30] as a zero-shot verifier: render each candidate from six viewpoints, refine each view, and measure refinement magnitude via LPIPS [33]. We then add a learned reranker and combine both

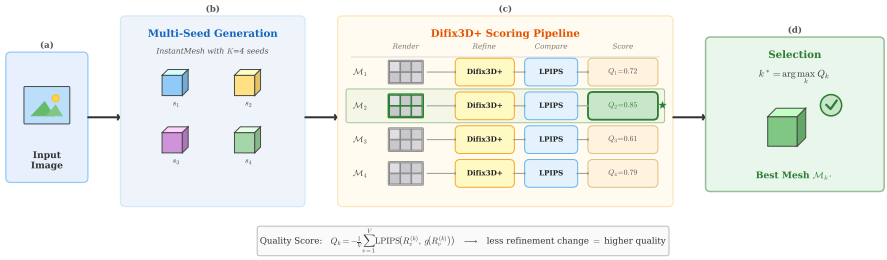

Fig. 2: SeedSelect pipeline. Given a single input image, we generate K candidate 3D reconstructions using different random seeds (Sec. 3.2). Each candidate is rendered from 6 canonical viewpoints and scored by measuring how much Difix3D+ modifies each rendering (Sec. 3.3). The candidate with the smallest average refinement gap is selected as the final output (Sec. 3.4).

via two-stage hybrid selection (coarse zero-shot filtering, fine learned reranking, risk-aware utility).

Our contributions are:

- We bring *inference-time compute scaling* to single-image 3D reconstruction and quantify the seed lottery phenomenon, where seed variation creates substantial exploitable quality headroom.
- We propose a unified test-time verification framework: a zero-shot multi-view verifier and a two-stage hybrid verifier that combines zero-shot robustness with learned precision.
- We show strong quality-cost trade-offs: on GSO-300, hybrid closes 42.3% oracle gap at substantially lower learned inference cost; on OmniObject3D-100, hybrid closes 52.3% and outperforms both zero-shot and learned-only under domain shift.
- We validate cross-dataset generalization and quantitative backbone transfer to LGM, provide qualitative transfer evidence on Wonder3D, and offer practical guidance on retention ratio α and risk parameter λ for deployment.

2 Related Work

Single-Image 3D Reconstruction. Recent methods combine multi-view diffusion models with feed-forward reconstruction to produce 3D assets from single images in seconds. Zero123 [7] first demonstrated novel-view synthesis by fine-tuning Stable Diffusion [16] on camera-conditioned pairs. Zero123++ [18] extends this to generate six views simultaneously, while MVDream [19] and SyncDreamer [8] enforce multi-view consistency via shared attention. On the reconstruction side, Large Reconstruction Models [3, 23, 26] use transformers to predict triplane representations, and InstantMesh [28] combines Zero123++

with FlexiCubes [17] mesh extraction. LGM [22] and GRM [29] use 3D Gaussian Splatting [4] for efficient reconstruction. A key architectural distinction is that pipelines with a stochastic diffusion stage (e.g., InstantMesh, LGM) exhibit seed-dependent quality variation, while fully deterministic models (e.g., TripoSR [23]) do not. Our work exploits this variation to improve output quality through inference-time selection.

Inference-Time Compute Scaling. The paradigm of allocating additional compute at test time has proven remarkably effective across domains. In language modeling, best-of-N sampling with reward models consistently improves output quality [21], and Snell *et al.* [20] show that scaling test-time compute can be more efficient than scaling model parameters. Ma *et al.* [11] recently extended this paradigm to 2D diffusion models, demonstrating inference-time scaling for image generation at CVPR 2025. In a related direction, Park *et al.* [12] show that seed selection improves text-to-image generation quality, though their approach operates in 2D with global (non-per-input) seed selection. In protein structure prediction, AlphaFold2 generates multiple predictions and selects via confidence scores. To our knowledge, this is the first systematic study of inference-time compute scaling via per-input seed verification for *single-image 3D reconstruction*. Unlike the 2D approaches of Ma *et al.* and Park *et al.*, our setting requires a quality verifier that operates on 3D outputs without ground truth—a fundamentally more challenging problem that we solve through multi-view proxy scoring.

Quality Assessment for 3D Generation. Evaluating generated 3D content typically requires ground-truth references via Chamfer Distance [10] or F-Score. Reference-free 3D quality metrics have received limited attention. Recent work has begun to address this: MATE-3D [5] proposes masking-and-tracking experts for 3D quality assessment, MVReward [6] learns multi-view preference alignment, and DreamReward [32] trains reward models for text-to-3D. However, these learned metrics require training data and are designed for text-to-3D rather than image-to-3D reconstruction. Simpler metrics such as mesh face regularity and surface smoothness do not correlate with semantic accuracy, and CLIP [14] similarity between input and rendered images fails to capture fine-grained geometric quality, as we demonstrate in Sec. 4.3. Our proxy instead exploits a pre-trained 2D refinement model's behavior as an implicit, zero-shot quality signal—requiring no training or 3D supervision.

2D Image Refinement for 3D Content. Difix3D [31] and its successor Difix3D+ [30] (CVPR 2025 Best Paper Finalist) are 2D image refinement models designed to fix visual artifacts in rendered 3D content, such as multi-face artifacts, texture bleeding, and geometric distortions. While designed for post-processing, we repurpose Difix3D+ as a quality verifier: the magnitude of refinement applied to a rendering serves as a proxy for the quality of the underlying 3D geometry. This novel application transforms a refinement tool into a scoring function for inference-time selection.

3 Method

SeedSelect brings inference-time compute scaling to diffusion-based 3D reconstruction. Given a single input image I, our method generates K candidate meshes $\{M_1, \ldots, M_K\}$ using different random seeds, scores each candidate via a multi-view quality verifier, and selects the highest-scoring one as the final output. In the language of inference-time scaling, the seed sweep constitutes the *search algorithm* and Difix3D+ serves as the *verifier* [20]. Figure 2 illustrates the pipeline.

3.1 The Seed Lottery

Modern single-image 3D reconstruction pipelines such as InstantMesh [28] consist of two stages: (i) a multi-view diffusion model (*e.g.*, Zero123++ [18]) generates novel views from the input image, and (ii) a reconstruction model produces a 3D mesh from these views. The diffusion stage is inherently stochastic: different random seeds $s \in \mathcal{S}$ lead to different noise realizations in the denoising process, producing different multi-view images and consequently different 3D reconstructions.

Let $f(I, s)$ denote the full reconstruction pipeline applied to input I with seed s, producing mesh M_s. Let $\mathrm{CD}(M_s, M^*)$ denote the Chamfer Distance between M_s and ground-truth mesh M^*. We observe empirically that

$$\frac{\max_{s\in\mathcal{S}} \mathrm{CD}(f(I,s), M^*)}{\min_{s\in\mathcal{S}} \mathrm{CD}(f(I,s), M^*)} \gg 1 \tag{1}$$

with ratios exceeding 3× for individual objects. Across our evaluation benchmarks, selecting the oracle-best seed from K=4 candidates reduces mean Chamfer Distance by 6.6% (GSO) and 11.9% (OmniObject3D), demonstrating that substantial quality improvements are available within the existing stochastic sampling distribution, *at no additional training cost*.

This motivates a generate-then-verify strategy: generate K candidates and use a verifier to select the best one.

Theoretical Motivation. Order statistics predicts that the expected improvement of the best-of-K over a single draw scales as $\sigma \cdot \Phi^{-1}(K/(K+1))$, where σ is the quality variance and Φ^{-1} is the normal quantile function [1]. This predicts a concave, monotonically increasing relationship with K—consistent with our empirical results (Tab. 10). Even an imperfect verifier suffices: if pairwise ranking accuracy exceeds 50% (random chance), the expected quality of the selected candidate exceeds the single-draw expectation.

3.2 Multi-Seed Candidate Generation

Given input image I and K random seeds $\{s_1, \ldots, s_K\}$, we generate K candidate meshes:

$$M_k = f(I, s_k), \quad k = 1, \ldots, K \tag{2}$$

In our experiments, we use InstantMesh [28] as the primary backbone pipeline f, with LGM [22] for quantitative backbone-transfer evaluation and Wonder3D [9] for qualitative transfer checks (Sec. 4.6). The generation cost scales linearly with K. We evaluate $K \in \{2, 4, 8, 16\}$.

3.3 Multi-View Quality Verifier

Intuition. A high-quality 3D reconstruction should produce clean, artifact-free renderings from all viewpoints. Conversely, geometric defects—holes, protrusions, self-intersections, missing parts—produce visual artifacts in rendered views that a 2D image refinement model would attempt to fix. We hypothesize that *the amount of refinement a 2D model applies to rendered views is inversely correlated with 3D quality*: better meshes produce renderings that need less fixing.

Multi-View Rendering. For each candidate mesh M_k, we render RGB images from V canonical viewpoints:

$$\mathcal{R}_k = \{R_k^{(v)} = \pi(M_k, c_v) \mid v = 1, \ldots, V\} \tag{3}$$

where π denotes the rendering function and c_v are camera parameters. We use V=6 viewpoints positioned at the front, back, left, right, top, and front-right of the object, providing comprehensive coverage of the surface.

Difix3D+ as Zero-Shot Verifier. We employ Difix3D+ [30], a pre-trained model designed to refine multi-view renderings of generated 3D content. Given a rendered view R, Difix3D+ produces a refined version $\tilde{R} = g(R)$ that attempts to correct visual artifacts while preserving the overall structure. The key property we exploit is that Difix3D+ applies *larger modifications to lower-quality renderings.*

We quantify the magnitude of refinement using LPIPS [33]:

$$\delta_k^{(v)} = \text{LPIPS}(R_k^{(v)}, g(R_k^{(v)})) \tag{4}$$

A smaller $\delta_k^{(v)}$ indicates that Difix3D+ found less to correct in view v of candidate k, suggesting higher quality.

Multi-View Score Aggregation. Single-view scores are noisy because defects may be visible from some viewpoints but not others, and Difix3D+ may apply view-specific changes unrelated to 3D quality. To obtain a robust quality estimate, we aggregate across all viewpoints:

$$Q_k = -\frac{1}{V} \sum_{v=1}^{V} \delta_k^{(v)} \tag{5}$$

The negation ensures that higher scores indicate better quality (less refinement needed). This multi-view averaging is essential: in our experiments, single-view scoring achieves only p=0.06 on GSO-300, while 6-view averaging reaches p=0.014 (Sec. 4.5).

3.4 Candidate Selection

The zero-shot output is the candidate with the highest multi-view quality score:

$$M^* = M_{k^*}, \quad k^* = \arg\max_k Q_k \tag{6}$$

3.5 Hybrid Verification

Zero-shot scoring is robust but limited by proxy fidelity, while learned verifiers can be more accurate but suffer from domain shift and higher compute. We therefore use a two-stage hybrid selector.

Stage 1: Zero-shot coarse filtering. Given K candidates, we keep the top-K' subset by zero-shot score:

$$K' = \max(2, \lfloor \alpha K \rfloor), \qquad \mathcal{C} = \mathrm{TopK}(Q, K') \tag{7}$$

where $\alpha \in [0.3, 0.5]$ controls retained candidates.

Stage 2: Learned reranking. For each candidate in $\mathcal{C}$, we build a hybrid feature vector

$$\phi_k = [\Delta_k^{\mathrm{LPIPS}}, \Delta_k^{\mathrm{CLIP}}] \in \mathbb{R}^{774} \tag{8}$$

where $\Delta_k^{\mathrm{LPIPS}} \in \mathbb{R}^6$ are per-view refinement magnitudes and $\Delta_k^{\mathrm{CLIP}} \in \mathbb{R}^{768}$ captures semantic difference between input image and rendered candidate views. A lightweight MLP verifier predicts a quality score

$$s_k = f_\theta(\phi_k), \quad k \in \mathcal{C}. \tag{9}$$

Stage 3: Risk-aware utility. To reduce brittle picks from uncertain predictions, we estimate uncertainty via MC dropout and rank by

$$u_k = \mathbb{E}[s_k] - \lambda \cdot \mathrm{Std}[s_k], \quad \lambda \in [0.1, 0.5], \tag{10}$$

then select

$$M^\dagger = \arg\max_{k \in \mathcal{C}} u_k. \tag{11}$$

Complexity. If $C_{\mathrm{LPIPS}}, C_{\mathrm{CLIP}}, C_{\mathrm{MLP}}$ denote per-candidate costs, learned-only reranking costs

$$K(C_{\mathrm{LPIPS}} + C_{\mathrm{CLIP}} + C_{\mathrm{MLP}}), \tag{12}$$

while hybrid costs

$$KC_{\mathrm{LPIPS}} + K'(C_{\mathrm{CLIP}} + C_{\mathrm{MLP}}). \tag{13}$$

Thus hybrid reduces expensive CLIP/MLP passes by roughly a factor of $K'/K \approx \alpha$.

Computational Overhead. In our implementation with K=4 (GSO) and K=8 (Omni), zero-shot scoring remains the fastest option; learned-only is the most expensive; hybrid provides a practical middle point with substantially

better quality-cost trade-off. Cost values in the experiments report verifier latency under one fixed runtime configuration shared by all methods (same hardware/software stack and inference settings), measured after candidate meshes and rendered views are available.

Verifier Robustness. A concern with generate-and-verify approaches is *verifier hacking*—the search algorithm exploiting verifier blind spots at large K. Our experiments show monotonic gains for zero-shot scaling and stable hybrid behavior across domains, with risk-aware selection further reducing unstable picks under distribution shift.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on two standard benchmarks: Google Scanned Objects (GSO) [2] with 300 randomly selected objects, and OmniObject3D [27] with 100 objects from 20 categories. Both datasets provide high-quality ground-truth 3D meshes.

Backbones. We evaluate SeedSelect on three diffusion-based 3D reconstruction pipelines spanning diverse architectures and 3D representations: (i) **InstantMesh** [28], which uses Zero123++ [18] for multi-view diffusion and FlexiCubes [17] for mesh extraction; (ii) **LGM** [22], which uses ImageDream [25] for multi-view diffusion and produces 3D Gaussian Splatting [4] output; (iii) **Wonder3D** [9], which uses a cross-domain multi-view diffusion model for joint color and normal generation, followed by NeuS [24] mesh reconstruction. All models are used off-the-shelf without modification. The primary evaluation uses InstantMesh; LGM is used for quantitative hybrid transfer evaluation, and Wonder3D is evaluated with GT-CD zero-shot/learned/hybrid comparison on a canonical GSO-150 subset (Sec. 4.6).

Metrics. We report Chamfer Distance (CD, $\times 10^{-2}$, $\downarrow$) as the primary 3D geometric metric, computed from 16K uniformly sampled surface points after alignment and normalization to $[-1, 1]^3$. We also report the following selection-quality metrics: *Improvement* (%), the relative CD reduction vs. the default seed; *Gap Closed* (%), the fraction of the oracle improvement achieved; *Oracle Match* (%), how often the method picks the best seed; *Worst Pick* (%), how often it picks the worst; and p-value from the Wilcoxon signed-rank test.

Baselines. We compare SeedSelect against: (i) **Default**: single-run with seed 42; (ii) **Random**: randomly select among K candidates; (iii) **PSNR**: re-render front view and compare against input image via PSNR; (iv) **Consensus**: pick the mesh closest to all others via pairwise CD; (v) **Geometric**: score by mesh face regularity and watertightness; (vi) **Smoothness**: score by surface normal consistency; (vii) **Difix3D+ (front)**: single front-view Difix3D+ score; (viii) **Oracle**: select the seed with lowest ground-truth CD (upper bound).

Table 1: Seed sensitivity in InstantMesh. Quality varies substantially across random seeds. Oracle best-of-K reveals large untapped potential.

	GSO (300 objects)		OmniObject3D (100 objects)	
	CD (↓)	Improv. (%)	CD (↓)	Improv. (%)
Default (seed 42)	0.365	—	0.684	—
Seed mean (K=4)	0.366	−0.3	0.668	+2.3
Oracle best-of-4	0.341	+6.6	0.603	+11.9
Max / Min CD ratio	up to 3.3×		up to 4.1×	

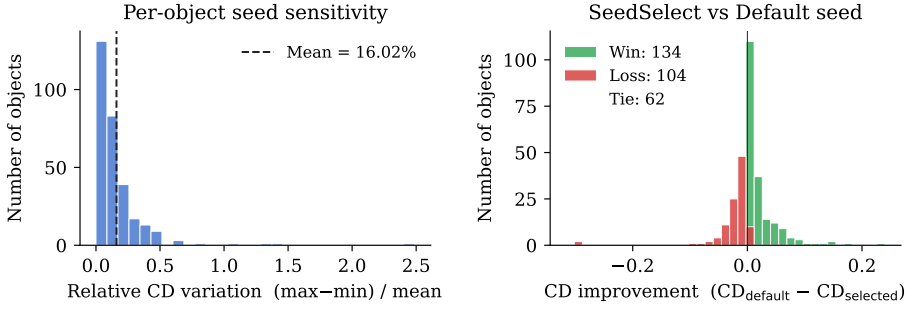

Fig. 3: Seed sensitivity distribution (GSO-300). *Left*: relative CD variation across 4 seeds per object. Most objects show 10–40% variation. *Right*: per-object improvement from SeedSelect vs. default. Green bars indicate objects that improve, red bars indicate objects that degrade.

4.2 Seed Sensitivity Analysis

Table 1 quantifies the seed lottery. On GSO, oracle best-of-4 achieves 6.6% lower CD than the default seed; on OmniObject3D the gap widens to 11.9%. The per-seed mean CD is comparable to the default, confirming that no single seed is universally superior—the benefit comes from per-object selection. Individual objects exhibit extreme variation (up to 3.3×).

4.3 Main Results

Table 2 compares all scoring methods on GSO-300. SeedSelect is the *only* method achieving statistically significant improvement (p=0.014), closing 22.1% of the oracle gap. Standard mesh quality metrics perform at random-selection level, and PSNR-based scoring fails due to camera/lighting mismatches. Notably, CLIP and DINOv2 similarity both fail (−0.3%, −0.4%, p>0.35), demonstrating that semantic similarity does not correlate with 3D geometric accuracy. Single-view Difix3D+ shows a trend (p=0.06) but fails to reach significance, demonstrating the importance of multi-view aggregation.

Table 2: Scoring method comparison on GSO-300 (K=4). Only SeedSelect (multi-view Difix3D+) achieves statistically significant improvement. All other methods—including CLIP and DINOv2 similarity—fail to outperform the default seed.

Method	CD (↓)	Improv. (%)	Gap (%)	Oracle (%)	Worst (%)	p-val
Default (seed 42)	0.365	—	—	—	—	—
Random	0.366	−0.3	—	25.0	25.0	—
PSNR	0.366	−0.3	—	26.0	24.7	0.91
Consensus	0.364	+0.2	3.2	26.3	24.3	0.82
Geometric	0.367	−0.5	—	25.3	27.3	0.76
Smoothness	0.366	−0.3	—	24.3	26.0	0.72
CLIP (ViT-B/32)	0.366	−0.3	—	23.3	26.0	0.42
DINOv2	0.366	−0.4	—	23.3	23.3	0.36
Difix3D+ (front)	0.363	+0.6	8.7	33.3	19.7	0.06
SeedSelect (Ours)	**0.360**	**+1.5**	**22.1**	**37.0**	**18.0**	**0.014**
Oracle	0.341	+6.6	100.0	100.0	0.0	—

Table 3: Cross-dataset validation (K=4). SeedSelect generalizes across benchmarks, with particularly strong results on OmniObject3D.

	GSO (300 objects)			OmniObject3D (100 objects)		
Method	Improv. (%)	Gap (%)	p-val	Improv. (%)	Gap (%)	p-val
Default	—	—	—	—	—	—
Difix3D+ (front)	+0.6	8.7	0.060	+5.0	42.0	3.3×10^{-4}
SeedSelect	**+1.5**	**22.1**	**0.014**	**+5.8**	**48.9**	$\mathbf{1.7 \times 10^{-4}}$
Oracle	+6.6	100.0	—	+11.9	100.0	—

Pairwise Ranking Accuracy. While the global Spearman correlation between proxy and GT quality is modest (ρ=−0.149), SeedSelect only needs to rank K candidates *per object*—a much easier task. We evaluate this via pairwise ranking accuracy: for each pair of candidates, does the proxy correctly identify the better one? Table 4 shows that SeedSelect achieves 55.3% on GSO-300 and 60.2% on OmniObject3D, significantly above the 50% random baseline. All other methods—CLIP, DINOv2, PSNR—fail to exceed random chance. Multi-view aggregation improves pairwise accuracy from 53.9% (front only) to 55.3% (6-view mean).

Table 3 shows that SeedSelect generalizes across datasets. On OmniObject3D, the improvement is even larger: 5.8% CD reduction, closing 48.9% of the oracle gap with p=1.7×10^{-4}. The stronger results on OmniObject3D likely reflect higher seed variance for the more diverse and challenging objects in this dataset.

Table 4: Pairwise ranking accuracy. Fraction of candidate pairs where the proxy correctly identifies the better reconstruction. SeedSelect is the only method exceeding random chance (50%).

Method	GSO-300 K=4	GSO-300 K=8	OmniObject3D K=4+
Random	50.0	50.0	50.0
PSNR	51.7	50.5	40.8
CLIP (ViT-B/32)	49.5	—	—
DINOv2	49.2	—	—
Difix3D+ (front)	53.9	—	56.8
SeedSelect (Ours)	**55.3**	**55.1**	**60.2**

Table 5: Hybrid verification results. Two-stage hybrid selection improves over zero-shot across domains while reducing cost versus learned-only inference.

Method	GSO-300			OmniObject3D-100		
	Gap (%)	Win (%)	Cost (ms)	Gap (%)	Win (%)	Cost (ms)
Zero-shot	22.14	44.67	40.0	43.68	62.00	80.0
Learned-only	68.86	58.00	444.0	41.96	60.00	888.0
Hybrid (best)	**42.35**	**50.00**	161.2	**52.32**	**64.00**	322.4

4.4 Hybrid Verification Results

In-domain vs out-of-domain behavior. On GSO-300, learned-only achieves the highest absolute gap-closed performance, while hybrid recovers a large fraction of that gain at substantially lower cost. On OmniObject3D-100, hybrid outperforms both zero-shot and learned-only, indicating stronger robustness under domain shift. This matches our hypothesis: zero-shot verification provides stable coarse filtering, while learned reranking contributes precision when features are reliable.

Retention-ratio ablation. Table 6 shows that α=0.3 is best or tied-best in gap-closed while minimizing verifier cost. Increasing α mostly increases cost, with limited in-domain gains and weaker OOD performance in our current runs.

Cost-normalized comparison. Table 8 reports gap-per-100ms under a fixed verifier runtime setting. Zero-shot is the most compute-efficient option, while learned-only has the weakest efficiency due to heavy per-candidate features. Hybrid improves efficiency over learned-only by 1.69× on GSO and 3.43× on Omni, while still improving absolute OOD quality over zero-shot (52.32% vs.

Table 6: Retention-ratio ablation ($\lambda \in \{0.1, 0.3, 0.5\}$, best shown per α). Lower α reduces cost; moderate α balances recall and reranking quality.

	GSO-300			OmniObject3D-100		
α	Gap (%)	Win (%)	Cost (ms)	Gap (%)	Win (%)	Cost (ms)
0.3	42.35	50.00	161.2	52.32	64.00	322.4
0.4	42.35	50.00	201.6	47.35	61.00	403.2
0.5	42.35	50.00	242.0	43.95	62.00	484.0

Table 7: Hybrid verification across backbones and domains. Gap closed is measured w.r.t. default seed (lower CD is better). Hybrid is strong on InstantMesh/LGM but does not outperform zero-shot on Wonder3D under the canonical protocol; significance varies by backbone/domain.

Domain	Zero-shot Gap (%)	Learned Gap (%)	Hybrid Gap (%)	Hybrid (α, λ)	Hybrid p-val
InstantMesh (GSO-300)	22.14	68.86	42.35	(0.3, 0.1)	1.02×10^{-7}
InstantMesh (Omni-100, OOD)	43.68	41.96	52.32	(0.3, 0.1)	5.59×10^{-6}
LGM (GSO-300)	-17.74	80.14	35.86	(0.3, 0.3)	1.50×10^{-5}
Wonder3D (GSO-150)	3.69	-9.86	2.01	(0.3, 0.3)	5.68×10^{-1}

43.68% gap closed on Omni). We report these efficiency values as normalized verifier-cost estimates from component-level profiling rather than end-to-end wall-clock latency, and treat them as relative comparison indicators. As a practical sanity check, on an RTX A6000 with precomputed candidate descriptors, measured selection-stage median latency is 0.0007/0.3233/2.4524 ms per object (zero-shot/learned/hybrid) on GSO-300 and 0.0011/0.3333/2.4462 ms on OmniObject3D-100.

Stronger learned-baseline check. Table 9 evaluates two additional learned rerankers that directly regress CD from candidate features. Despite moderate global correlation (0.57–0.58), both regressors fail to improve test-time selection quality on GSO. In contrast, our ranking-oriented learned verifier is strongly positive, highlighting that the objective design (ranking under per-object competition) matters as much as model capacity.

Cross-backbone hybrid behavior. Table 7 extends hybrid verification beyond the primary InstantMesh setting. On LGM, zero-shot scoring alone is unreliable (negative gap-closed), while learned and hybrid reranking recover strong gains under the same evaluation protocol. On Wonder3D (canonical GSO-150), zero-shot and hybrid are marginally positive while learned-only is negative, and all methods remain statistically non-significant at current sample size. This highlights backbone-dependent verifier fidelity and the current limitation of learned transfer on Wonder3D.

Table 8: Verifier cost-efficiency. Gap-per-100ms is computed as gap-closed divided by verifier latency. Zero-shot is most compute-efficient, while hybrid provides a stronger quality–efficiency balance than learned-only.

Method	GSO-300			OmniObject3D-100		
	Gap (%)	Cost (ms)	Gap/100ms	Gap (%)	Cost (ms)	Gap/100ms
Zero-shot	22.14	40.0	55.35	43.68	80.0	54.60
Learned-only	68.86	444.0	15.51	41.96	888.0	4.73
Hybrid (best)	42.35	161.2	26.27	52.32	322.4	16.23

Table 9: Naive learned rerankers on GSO-300 (K=4). Direct CD regression is insufficient for per-object seed ranking, even with non-trivial global correlation. Our learned-only verifier (ranking-oriented objective) is substantially stronger.

Method	Params	Corr. (↑)	Improv. (%)	Gap (%)	Win (%)	p-val
MLP-6D (LPIPS only, CD regression)	9.2K	0.583	−0.01	−0.17	34.3	0.95
MLP-774D (CLIP+LPIPS, CD regression)	562.9K	0.570	−0.39	−5.90	33.0	0.11
Learned-only (ours, ranking objective)	562.9K	—	**+4.53**	**68.86**	**58.0**	1.95×10^{-17}

4.5 Scaling with K

Table 10 examines how SeedSelect scales with K. SeedSelect improvements increase monotonically from +1.9% (K=2) to +5.4% (K=16), with statistical significance maintained throughout (p<0.05 for all K). At K=16, the oracle gap reaches 14.3%, showing that the seed lottery provides an ever-larger pool of quality to exploit. On the full GSO-300, K=8 achieves 2.3% improvement with p=9.6×10^{-4}. The scaling trend holds across datasets: on OmniObject3D, increasing from K=4 to K=8 improves the absolute CD reduction from 5.8% to 6.3% while the worst-pick rate drops from 21% to 9%, demonstrating that more candidates benefit both quality and robustness. Figure 5 visualizes this scaling behavior alongside the theoretical order-statistics prediction.

Ablation Studies Table 11 demonstrates the critical role of multi-view aggregation (*left*). A single front view provides weak signal (p=0.06), but performance improves sharply from 3 views onward, confirming that geometric defects manifest differently across viewpoints. Mean aggregation (*right*) achieves the highest gap closed (22.1%), while max aggregation has the lowest p-value (0.009) due to sensitivity to outlier defects. Intermediate-view rows are computed from a fixed candidate pool with cached multi-view scores under the same protocol to isolate the view-count effect.

Figure 6 shows qualitative examples. Default seeds often produce geometric artifacts (missing parts, distorted surfaces), while SeedSelect reliably identifies higher-quality candidates approaching oracle quality.

Proxy–Quality Correlation. The multi-view proxy score shows moderate but significant correlation with ground-truth CD (Spearman ρ=−0.149, p=2.9 ×

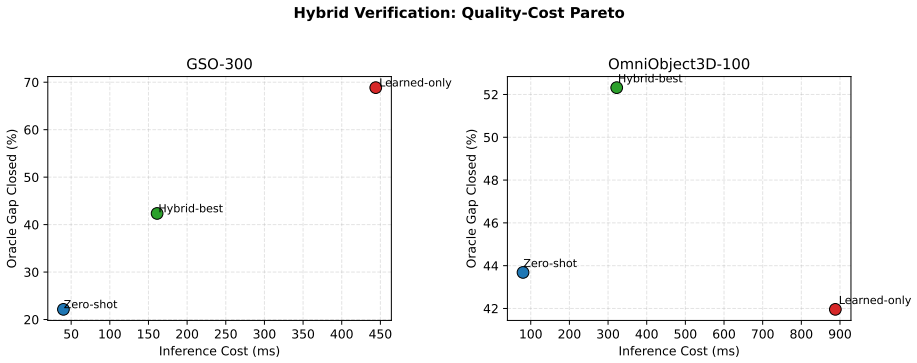

Fig. 4: Quality-cost Pareto of zero-shot, learned-only, and hybrid verification. Hybrid provides the strongest trade-off region: it consistently improves over zero-shot while using much less compute than learned-only.

10^{-7}). While global correlation is modest, the pairwise ranking accuracy of 55–60% (Tab. 4) confirms the proxy reliably distinguishes better from worse candidates at the per-object level.

Failure Analysis. SeedSelect degrades quality for 35% of objects, but the damage is typically mild: 62% of failures cause <5% CD increase, with a median degradation of only 4.0%. Success rate is consistent across objects regardless of their seed sensitivity (CD range), indicating the proxy does not systematically fail on particular object types. Over the full evaluation, the expected value of selection is strongly positive.

Selection vs. Refinement. A natural question is why not refine the default reconstruction using Difix3D+ directly. However, Difix3D+ modifies *rendered pixel values*, not 3D geometry. Applying it to rendered views improves appearance but leaves the mesh (and its Chamfer Distance) unchanged. Obtaining geometric improvement from refined 2D views would require full 3D reconstruction from the refined images—a substantially more complex pipeline. SeedSelect achieves genuine 3D improvement through selection alone, using Difix3D+ only as a lightweight scoring signal.

4.6 Cross-Backbone Generalization

Backbone transfer is a key stress test for verifier robustness. Beyond InstantMesh, we evaluate the same hybrid pipeline on LGM [22] (ImageDream + 3DGS) using per-seed CD. The zero-shot proxy degrades on LGM, but learned and hybrid reranking remain significantly better than the default seed (Tab. 7), indicating that hybrid verification can absorb backbone-specific proxy mismatch.

For Wonder3D [9] (diffusion + NeuS), we complete a full GT-based comparison on canonical GSO-150 (Tab. 12). Relative to default seed, zero-shot improves mean CD by +0.09%, learned-only degrades by -0.23%, and hybrid gives +0.05%; all remain statistically non-significant (p>0.45). This indicates

Table 10: Scaling with number of candidates K. Both oracle and SeedSelect improvements increase monotonically with K, with statistical significance maintained up to K=16.

K	Dataset	Oracle	Ours	Gap	Worst	p-val
50-object subset						
2	GSO-50	+5.4	+1.9	30.3	—	4.5×10^{-2}
4	GSO-50	+9.0	+3.2	35.0	—	7.0×10^{-3}
8	GSO-50	+11.8	+4.8	40.5	6.0	7.2×10^{-4}
16	GSO-50	+14.3	+5.4	37.5	6.0	2.4×10^{-3}
Full evaluation						
4	GSO-300	+6.6	+1.5	22.1	18.0	1.4×10^{-2}
8	GSO-300	+8.3	+2.3	27.2	17.0	9.6×10^{-4}
Cross-dataset (OmniObject3D, 100 objects)						
4	Omni	+11.9	+5.8	48.9	21.0	2.6×10^{-4}
8	Omni	+14.4	+6.3	43.7	9.0	1.5×10^{-4}

Table 11: Ablation studies (GSO-300, K=4). *Left*: Multi-view aggregation is critical—performance improves sharply from 1 to 4+ views. *Right*: Mean aggregation outperforms alternatives.

Views V	Improv.	Gap	p	Aggregation	Improv.	Gap	p
1	+0.6	8.7	0.060	Front only	+0.6	8.7	0.060
2	+0.4	5.7	0.142	Max	+1.3	20.4	0.009
4	+1.4	21.9	0.005	Front-wt.	+1.0	15.1	0.017
6	**+1.5**	**22.1**	**0.014**	**Mean**	**+1.5**	**22.1**	**0.014**

Wonder3D remains a harder transfer regime than InstantMesh/LGM under the current verifier setup.

Runtime. Hybrid preserves the intended efficiency profile across settings: it is consistently cheaper than learned-only verification, while delivering substantially higher quality than zero-shot in challenging transfer cases (e.g., LGM). Reported cost (ms) is verifier-only latency under a fixed runtime setup shared across methods, excluding multi-seed candidate generation.

5 Conclusion

SeedSelect demonstrates inference-time compute scaling for 3D reconstruction through test-time verification. We show that diffusion seed stochasticity creates large quality headroom, then exploit it with (i) zero-shot multi-view verification and (ii) hybrid verification that combines zero-shot filtering with learned reranking. Across benchmarks, hybrid improves the quality-cost frontier: on GSO-300 it

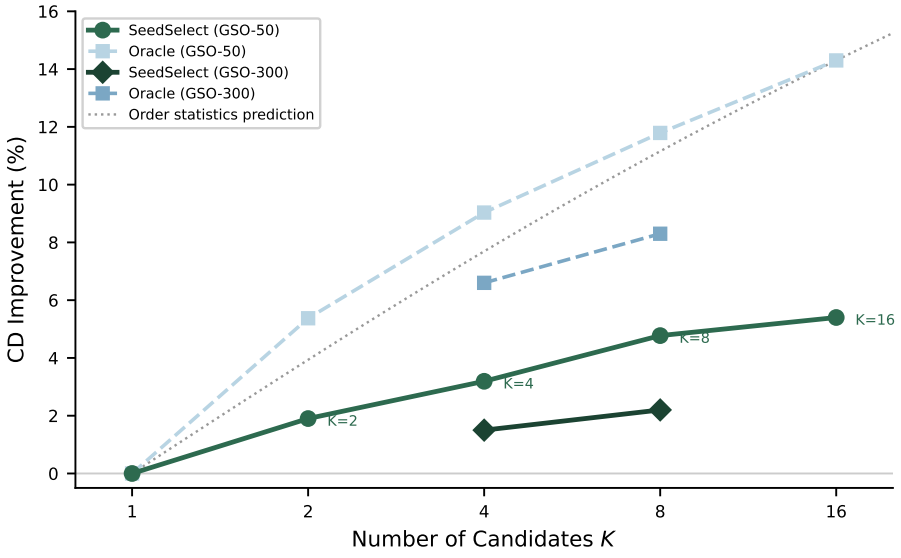

Fig. 5: Inference-time scaling curve. CD improvement increases monotonically with K, following the order-statistics prediction (dotted). SeedSelect consistently captures a significant fraction of the oracle improvement.

closes 42.3% oracle gap at substantially lower learned-only verification cost, and on OmniObject3D-100 it closes 52.3% while outperforming both zero-shot and learned-only under domain shift. These results indicate that robust coarse verification plus lightweight learned reranking is a practical path toward deployable test-time scaling for 3D.

Limitations. SeedSelect requires K forward passes, so generation cost still scales linearly with candidate count. Learned and hybrid verifiers depend on feature quality; domain shift can weaken learned-only ranking. Our current risk-aware term uses uncertainty from MC dropout, but richer uncertainty estimators may further improve robustness.

Future Work. Promising directions include pairwise ranking objectives for learned verifiers, adaptive α/K scheduling by object difficulty, and broader verifier transfer across backbones and datasets with full GT-based evaluation. SeedSelect is also complementary to post-hoc refinement: selecting stronger candidates first can improve the starting point for downstream optimization or refinement pipelines.

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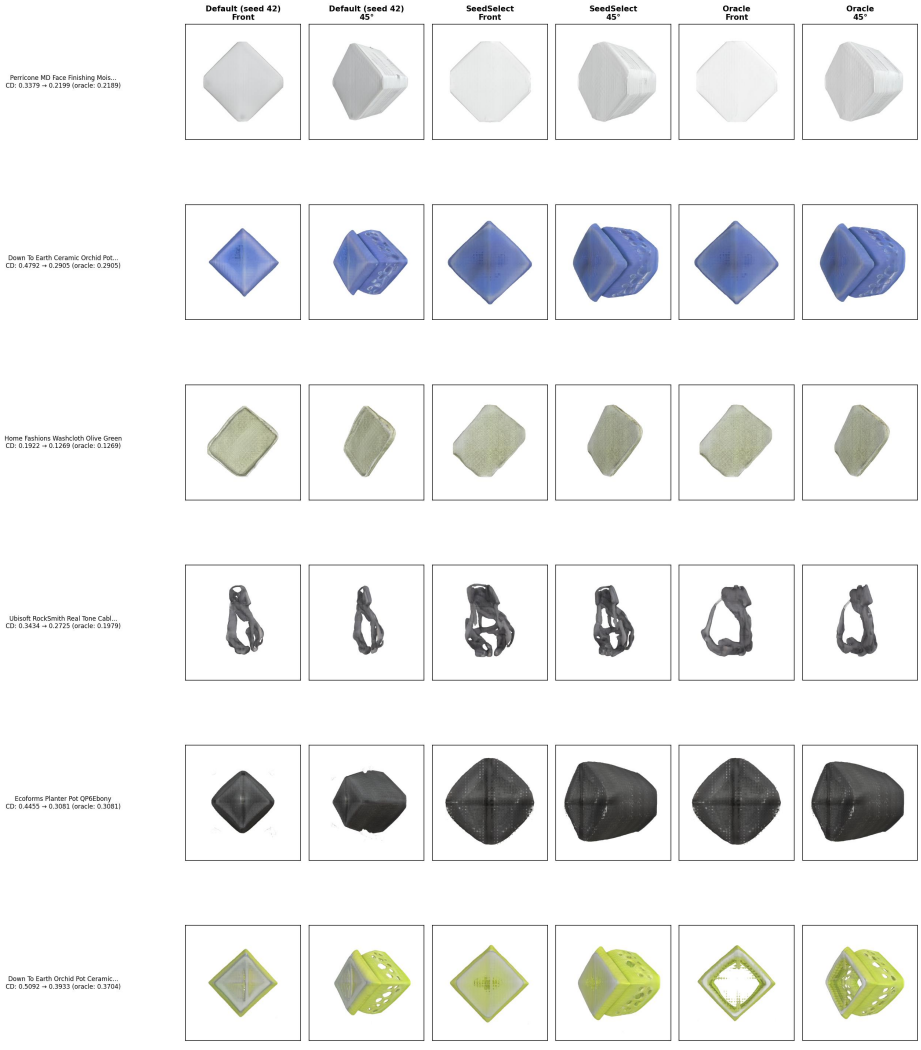

Fig. 6: Qualitative comparison. For six GSO objects, we show the default reconstruction (seed 42), the SeedSelect output, and the oracle-best reconstruction, each rendered from two viewpoints (front and 45°). CD values are annotated per row. SeedSelect consistently selects candidates with cleaner geometry, approaching oracle quality.

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Table 12: Wonder3D GT-CD evaluation (GSO-150, K=4, canonical locked protocol). Zero-shot and hybrid are marginally positive; learned-only is negative. None of the gains are statistically significant at this sample size.

Method	CD (↓)	Improv. (%)	Gap (%)	Win (%)	p-val
Default (seed 42)	0.2163	—	—	—	—
Zero-shot	**0.2161**	+0.09	**3.69**	38.0	0.458
Learned-only	0.2168	−0.23	−9.86	33.3	0.637
Hybrid (0.3, 0.3)	0.2162	+0.05	2.01	38.0	0.568
Oracle	0.2112	+2.35	100.0	—	—

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